MATH 241: Calculus III

Fall 2026

Lecture 22: Line Integrals

Background & Motivation

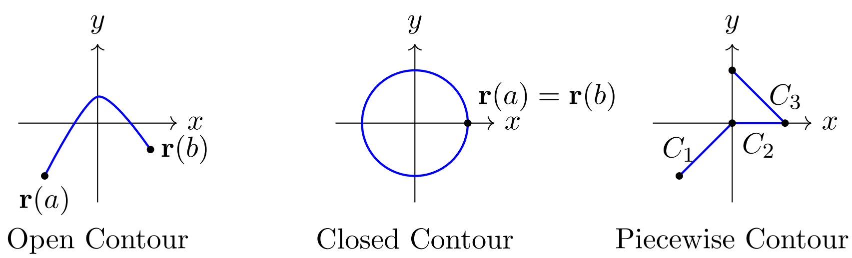
With vector fields defined, we now integrate them along curves. The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ computes the work done by a force along a path. For conservative fields, the Fundamental Theorem of Line Integrals gives us a shortcut: the integral depends only on the endpoints, not the path.

1. Contour Types

A **contour** (or **curve**) C is a path in $\mathbb{R}^2$ or $\mathbb{R}^3$ given by a parameterization $\mathbf{r}(t)$, $a \le t \le b$.

Definition 1: Types of Contours

- **Open** – endpoints $\mathbf{r}(a) \neq \mathbf{r}(b)$
- **Closed** – $\mathbf{r}(a) = \mathbf{r}(b)$ (loop); use notation $\oint_C$
- **Simple** – does not cross itself
- **Piecewise smooth** – finitely many smooth pieces joined end-to-end



Common Parameterizations

- Line segment from A to B – $\mathbf{r}(t) = (1-t)A + tB, \quad t \in [0, 1]$
- Circle $x^2 + y^2 = R^2$ – $\mathbf{r}(t) = \langle R\cos t,\, R\sin t\rangle, \quad t \in [0, 2\pi]$
- Graph $y = f(x)$, $a \le x \le b$ – $\mathbf{r}(t) = \langle t,\, f(t)\rangle, \quad t \in [a, b]$
- Helix – $\mathbf{r}(t) = \langle a\cos t,\, a\sin t,\, bt\rangle$

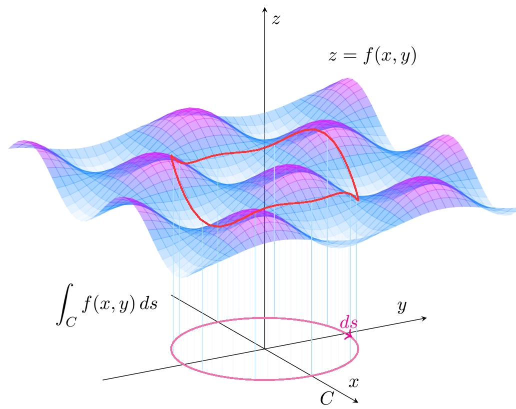
2. Scalar Line Integrals

Definition 2: Scalar Line Integral

For a scalar function f along a curve C parameterized by $\mathbf{r}(t)$:

$$\int_C f\,ds = \underline{\hspace{6cm}}$$

– $ds = \|\mathbf{r}'(t)\|\,dt$ is the **arc length element**
– **Interpretation:** area of the "curtain" under f along C
– Direction does not matter: $\int_C f\,ds = \int_{-C} f\,ds$



Example 1: Scalar Line Integral Along a Line Segment

Evaluate $\displaystyle\int_C (x + 2y)\,ds$ where C is the segment from $(0, 0)$ to $(3, 4)$.

3. Vector Field Line Integrals

Definition 3: Line Integral of a Vector Field

For a vector field $\mathbf{F}$ along a curve C:

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \underline{\hspace{6cm}}$$

– **Interpretation:** $\underline{\hspace{7cm}}$
– Reversing direction **negates** the integral: $\int_{-C} \mathbf{F}\cdot d\mathbf{r} = -\int_C \mathbf{F}\cdot d\mathbf{r}$

– Equivalent notation: $\int_C P\,dx + Q\,dy + R\,dz$

Example 2: Work Around a Circle

Compute $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r}$ for $\mathbf{F} = \langle -y,\, x\rangle$ along the circle $x^2 + y^2 = 4$ (CCW).

4. Fundamental Theorem of Line Integrals (FTOLI)

Theorem 1: Fundamental Theorem of Line Integrals

If $\mathbf{F} = \nabla f$ (conservative) and C goes from A to B:

$$\boxed{\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)}$$

Consequences:

- The integral depends only on ________________ – **path independence**
- For any closed curve: $\oint_C \nabla f \cdot d\mathbf{r} =$ ________________
- If $\oint_C \mathbf{F}\cdot d\mathbf{r} \neq 0$ for some closed C, then $\mathbf{F}$ is **not** conservative

Example 3: Applying FTOLI

Evaluate $\displaystyle\int_C \langle ye^{xy},\, xe^{xy}\rangle \cdot d\mathbf{r}$ from $(0,0)$ to $(1,2)$.

Example 4: Piecewise 3D Line Integral

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y^2,\, z^2,\, x^2 \rangle$ along the path $(0,0,0) \to (1,0,0) \to (1,1,0) \to (1,1,1)$.

Practice Problems

Problem 1. Evaluate $\int_C xy\, ds$ where C is the line segment from $(0,0)$ to $(1,1)$.

Problem 2. Evaluate $\int_C \langle y^2,\, 2xy \rangle \cdot d\mathbf{r}$ from $(0,0)$ to $(1,1)$ along $y = x^2$.

Problem 3. Evaluate $\oint_C \langle x^2,\, y^2 \rangle \cdot d\mathbf{r}$ where C is the unit circle. (*Hint: is the field conservative?*)

Problem 4. Evaluate $\int_C \langle 2xz + y,\ x + z^2,\ x^2 + 2yz \rangle \cdot d\mathbf{r}$ from $(0,0,0)$ to $(1,2,3)$ using FTOLI.
(*Hint: find the potential function first.*)

Problem 5. Evaluate $\int_C (x^2 + y^2)\, ds$ where C is the helix $\mathbf{r}(t) = \langle \cos t,\ \sin t,\ t \rangle$, $0 \leq t \leq 2\pi$.